Draft: The Carried Insertions at Complex Data

1 The Carried Insertions at Complex Data

The clause (5.21) of Definition 5.2 bounds the insertions on the whole space of the step, and not only at its real backgrounds. For the groups which a step makes this is part of what the count of that step delivers, in the form (5.22). For a group W which was made at an earlier step j and is carried to the step $k+1$ the question is of another kind, since its insertion is the function which was bounded at the step j, and it has now to be bounded on the space of the step $k+1$ and summed over the groups at a traversal of that step. The obvious bound fails. In the units of the fine lattice the space of the step $k+1$ lies in that of the step j, so that the supremum $\|Y_W\|_{(j)}$ of the step j bounds the insertion; but a traversal of the step $k+1$ consists of L^m traversals of the step j, $m \coloneqq k+1-j$, each with carried groups of its own, and the sum of these suprema is larger than the bound of the step j by the factor L^m. At the real backgrounds this factor is compensated by the contraction of the deviation (Section 6.9). At complex data there is no deviation to contract, and the compensation has to come from the size of the complex part: the radii of the space of the step $k+1$ are smaller than those of the step j by the factor L^{-m}, up to the ratio of the couplings, so that Schwarz's lemma on the larger space gains L^{-m} for each insertion. This cancels the number of the traversals and no more, and the sum over the generations $m = 1, \dots, k$ would still grow with k.

What removes the growth is gauge covariance at complex gauge transformations. The complex part of a configuration of the space of the step $k+1$ varies slowly, by the gradient condition of that space, and on a set which is small in the units of the scale $k+1$ it is therefore, to first order, a complex gauge transformation with a gauge function linear in the position. The dependence set of a carried group is such a set, of diameter δ_W of the order of $ML^{-m}\sigma(W)$ in these units. Removing the linear part by a gauge transformation which is the identity at the first site of the interval of W leaves a complex part smaller by the factor δ_W, and Schwarz's lemma then gains $L^{-2m}\sigma(W)$, which is summable over the generations; the factor $\sigma(W)$, and the groups whose dependence sets are not small, are paid by a sixteenth of the weight $e^{\nu\sigma(W)}$. This is the reason for the rate $\lambda_c < 1$ in (5.21). The insertions are covariant under gauge transformations with values in the compact group only, and the first step is to see that the covariance extends.

Throughout this section we work on the labelled term without large-field regions, we fix steps $1 \le j \le k$ and put $m \coloneqq k+1-j$, and we measure lengths in the units of the fine lattice, so that a traversal of the step n consists of L^n fine bonds and a cube of π_n has the side ML^n. We write $\mathfrak{g}$ for the Lie algebra of G, $\mathfrak{g}^c \coloneqq \mathfrak{g} \otimes \mathbb{C}$, and G^c for the complexification of G, the connected complex Lie group $G\exp(i\mathfrak{g})$ of matrices with the Lie algebra $\mathfrak{g}^c$; a representation π of G extends to a holomorphic representation of G^c. The norm $|\cdot|$ is that of Definition 5.2, on the matrices of the size of G, and it is invariant under the conjugation by the elements of G.

1.1 The Spaces of the Steps in Fine Units

For a configuration U with values in G and a field α on the fine bonds with values in $\mathfrak{g}^c$ we write $e^{i\alpha}U$ for the configuration $b \mapsto e^{i\alpha(b)}U(b)$, with values in G^c. For a bond $b = \langle x, x+e_\mu\rangle$ and a unit vector e, one of the $\pm e_\nu$, let $b+e \coloneqq \langle x+e, x+e+e_\mu\rangle$, and let the *covariant difference* of α be

$$(\delta_e\alpha)(b) \coloneqq U(x,x+e)\,\alpha(b+e)\,U(x,x+e)^{-1} - \alpha(b)\;, \tag{1.1}$$

where $U(x,x+e)$ is the value of U on the bond from x to $x+e$. For $n \geq 2$ let $\mathcal{S}_n(U)$ be the set of the pairs $(e^{i\alpha}U, \mathbf{J})$, with fields α and $\mathbf{J}$ on the fine bonds with values in $\mathfrak{g}^c$, such that for all bonds b, unit vectors e, and fine plaquettes p

$$\begin{gathered} |\alpha(b)| < \alpha_{1,n}L^{-n}\;, \qquad |(\delta_e\alpha)(b)| < \alpha_{1,n}L^{-2n}\;, \\ |(e^{i\alpha}U)(\partial p) - 1| < c_F\,\alpha_{0,n}L^{-2n}\;, \qquad |\mathbf{J}(b)| < \alpha_{0,n}L^{-3n}\;, \end{gathered} \tag{1.2}$$

with a constant $c_F \in (0,1]$ of the class; $\mathcal{S}_1(U)$ is the set of the configurations $e^{i\alpha}U$ with $|\alpha(b)| < \alpha_{1,1}L^{-1}$, without a variable $\mathbf{J}$. These are the conditions of Definition 5.2(iii) on the term without large-field regions. Indeed, in the units of the scale n the fine lattice has the spacing $\xi = L^{-n}$, the complex part is written $e^{i\xi A'}$, so that $\alpha = \xi A'$, and the conditions [2, (2.39), (2.34), (2.36), p. 261] on the top domain read $|A'| < \alpha_{1,n}$, $|\nabla_U^\xi A'| < \alpha_{1,n}$ with $\nabla_U^\xi = \xi^{-1}\delta$, $|\partial\mathbf{U} - 1| < \alpha_{0,n}\xi^2$, and $|\mathbf{J}| < \alpha_{0,n}$; a current is of the dimension of the third power of an inverse length [1, (1.8), p. 261], whence the factor L^{-3n} in fine units. For a set X of bonds we write $\mathcal{S}_n(U)|_X$ for the set defined by the same conditions on the bonds of X and on the plaquettes and differences which they contain. This set is open in the space of the pairs of a G^c-valued configuration and a $\mathfrak{g}^c$-valued field on X, because $\alpha \mapsto e^{i\alpha}U(b)$ maps a neighbourhood of 0 in $\mathfrak{g}^c$ biholomorphically onto a neighbourhood of $U(b)$ in G^c, and a gauge transformation v with values in G maps $\mathcal{S}_n(U)|_X$ onto $\mathcal{S}_n(U^v)|_X$, by the invariance of the norm.

We use the following properties of the class, for a group W of the deviation species made at the step j; it has one interval I_W, from a site x_W to a site x'_W, which lies in its domain X_W.

(a) The insertion Y_W is a function of the restriction of $(\mathbf{U},\mathbf{J})$ to the dependence set X_W^+ (of $\mathbf{U}$ alone for $j=1$). It is defined and analytic on the open set

$$\mathcal{D}_W \coloneqq \bigcup_U \mathcal{S}_j(U)|_{X_W^+}\;, \tag{1.3}$$

the union over the configurations U which are the trivial configuration, the background $U_j(V_j)$ of a block field V_j in the support of the test functions of the term, or a gauge transform of one of these by a G-valued gauge transformation. On $\mathcal{D}_W$ it is bounded by $\|Y_W\|_{(j)}$, and for G-valued gauge transformations v

$$Y_W(\mathbf{U}^v, R(v)\mathbf{J}) = \pi(v(x_W))\,Y_W(\mathbf{U},\mathbf{J})\,\pi(v(x'_W))^{-1}\;, \tag{1.4}$$

where $\mathbf{U}^v(b) = v(b_-)\mathbf{U}(b)v(b_+)^{-1}$ and $(R(v)\mathbf{J})(b) = v(b_-)\mathbf{J}(b)v(b_-)^{-1}$ [1, (1.10), p. 262], and π is the representation of the loop of W (Definition 5.2(iii) and (vii)). At a later step the insertion of W is the same function of the configuration and of the current, both in fine units.

(b) For V_{k+1} in the support of the step $k+1$ the background U_{k+1} is a background of the step j: with $V_i^* \coloneqq M^i(U_{k+1})$ for $j \le i \le k$, the block field V_i^* lies in the support of the step i (Supplement, Proposition S.8, by descending induction on i), and $U_i(V_i^*) = U_{k+1}$ by the uniqueness of the minimal orbit (Section 6.1). Moreover $|U_{k+1}(\partial p) - 1| < 2B_3\varepsilon_{k+1}L^{-2(k+1)}$ for every fine plaquette p [2, p. 268], and the pair of U_{k+1} and its current J_{k+1} is a point of $\mathcal{S}_{k+1}(U_{k+1})$.

(c) The domain X_W is a connected union of at most $u_\star\sigma(W)$ cubes of π_j, and I_W has at most $u_\star M\sigma(W)L^j$ fine bonds (Definition 5.2(vii)).

(d) The radii of two steps $n' < n$ satisfy $\alpha_{i,n} \le \varkappa_{n-n'}\,\alpha_{i,n'}$ for $i = 0, 1$, with $\varkappa_m \coloneqq (1+\beta_0)^2 m^{1/2}$, by (3.9) and (3.10), and $\alpha_{1,n} = \lambda_\alpha \alpha_{0,n}$ with $\lambda_\alpha \coloneqq C_1^\alpha / C_0^\alpha$, since $q_1 = q_0$ (Appendix A).

We assume

$$L \ge 20(1+\beta_0)^2\,, \tag{1.5}$$

so that $\varkappa_m L^{-m} \le (1+\beta_0)^2 L^{-1} \le \frac{1}{20}$ for every $m \ge 1$, because $m^{1/2}L^{-(m-1)} \le 1$ for $L \ge 2$.

Lemma 1.1 (The Space of a Later Step Lies in That of an Earlier One)**.** *Let U be a G-valued configuration. Then $\mathcal{S}_{k+1}(U) \subset \mathcal{S}_j(U)$.*

Proof. By (d) and (1.5), $\alpha_{i,k+1}L^{-(k+1)} \le \varkappa_m L^{-m}\alpha_{i,j}L^{-j} \le \alpha_{i,j}L^{-j}$, and the same holds with the powers $L^{-2(k+1)}$ and $L^{-3(k+1)}$, which are smaller. Each of the four radii of (1.2) at $n = k+1$ is therefore at most the corresponding radius at $n = j$, and for $j = 1$ only the first condition is posed. *q.e.d.*

1.2 Complex Gauge Transformations

Lemma 1.2 (Complexified Covariance)**.** *Let X be a finite set of bonds with the set S of its sites, $x, x' \in S$, $\mathcal{D}$ an open set of pairs $q = (\mathbf{U}, \mathbf{J})$ of a G^c-valued configuration and a $\mathfrak{g}^c$-valued field on X which is invariant under the G-valued gauge transformations, and F a holomorphic function on $\mathcal{D}$ with values in the linear maps of the space of a representation π of G, such that $F(q^v) = \pi(v(x))F(q)\pi(v(x'))^{-1}$ for every $q \in \mathcal{D}$ and every $v \in G^S$, where $q^v \coloneqq (\mathbf{U}^v, R(v)\mathbf{J})$. Let $q \in \mathcal{D}$ and let $t \mapsto u_t$, $t \in [0,1]$, be a continuous path in $(G^c)^S$ with $u_0 = 1$ and $q^{u_t} \in \mathcal{D}$ for every t. Then*

$$F(q^{u_1}) = \pi(u_1(x))\,F(q)\,\pi(u_1(x'))^{-1}\,. \tag{1.6}$$

Proof. The map $u \mapsto q^u$ is holomorphic on $(G^c)^S$, so that $\mathcal{U} \coloneqq \{u \in (G^c)^S \mid q^u \in \mathcal{D}\}$ is open, and it contains G^S by the invariance of $\mathcal{D}$. Since π is holomorphic on G^c, the function

$$\Psi(u) \coloneqq \pi(u(x))^{-1}F(q^u)\pi(u(x')) - F(q)$$

is holomorphic on $\mathcal{U}$, and it vanishes on G^S by hypothesis. In the chart $Z \mapsto \exp(Z)$ of $(G^c)^S$ at the identity, with $Z \in (\mathfrak{g}^c)^S = \mathfrak{g}^S \oplus i\mathfrak{g}^S$, the function $\Psi \circ \exp$ is holomorphic near 0 and vanishes on the real subspace $\mathfrak{g}^S$, whose complex span is the whole space. All its derivatives at 0 in the real directions vanish, hence, by complex linearity of the derivatives of a holomorphic function, all its derivatives at 0, and Ψ vanishes on a neighbourhood of the identity. By the identity theorem it vanishes on the connected component of $\mathcal{U}$ which contains the identity, and the path u_t lies in this component, because it lies in $\mathcal{U}$ and starts at the identity. Hence $\Psi(u_1) = 0$, which is (1.6). *q.e.d.*

We use the lemma for the insertion of W with the transport along its interval removed. Let $\mathbf{U}(I_W)$ be the ordered product of $\mathbf{U}$ along the fine bonds of I_W, from x_W to x'_W, an element of G^c, and

$$\hat{Y}_W(\mathbf{U},\mathbf{J}) \coloneqq Y_W(\mathbf{U},\mathbf{J})\,\pi(\mathbf{U}(I_W))^{-1}\ . \tag{1.7}$$

Since $\mathbf{U}^v(I_W) = v(x_W)\mathbf{U}(I_W)v(x'_W)^{-1}$, the function $\hat{Y}_W$ is holomorphic on $\mathcal{D}_W$ and satisfies (1.4) with x_W in the place of x'_W. By Lemma 1.2, for a path u_t as there with $u_1(x_W) = 1$,

$$\hat{Y}_W(\mathbf{U}^{u_1}, R(u_1)\mathbf{J}) = \hat{Y}_W(\mathbf{U},\mathbf{J})\ . \tag{1.8}$$

The gauge transformation which we shall use is built from the following function. Three properties of the norm enter, each with a constant which depends only on G and on the norm; we fix $\rho_0 \in (0,1]$ and $c_G \geq 1$ such that all three hold.

(N1) For $X, Q \in \mathfrak{g}^c$ with $|X| \leq \rho_0$, $|e^{iX}Qe^{-iX} - Q| \leq c_G|X||Q|$.

(N2) For $X, Y, Z, A \in \mathfrak{g}^c$ of norm less than ρ_0 the matrix $e^{iX}e^{iY}e^{iZ}e^{-i(X+A)}e^{-iZ}$ is within the radius of convergence of the logarithm about 1, and

$$\Phi(X,Y,Z,A) \coloneqq -i\log(e^{iX}e^{iY}e^{iZ}e^{-i(X+A)}e^{-iZ}) \tag{1.9}$$

is a holomorphic function with values in $\mathfrak{g}^c$; the function $\Psi \coloneqq \Phi - (Y - A)$ has all its first and second partial derivatives bounded by c_G in norm where the four arguments have norm at most $\rho_0/2$.

(N3) Let U be a G-valued configuration and α a field with $|\alpha(b)| \leq a_\circ \leq \rho_0$ and $|(\delta_e\alpha)(b)| \leq d_\circ \leq a_\circ$ on the bonds of a fine plaquette p. Then

$$|(e^{i\alpha}U)(\partial p) - 1| \leq c_G(|U(\partial p) - 1| + d_\circ + a_\circ^2)\ . \tag{1.10}$$

Such constants exist. (N1) holds because $Q \mapsto e^{iX}Qe^{-iX} - Q$ is holomorphic in X and vanishes at $X = 0$. In (N2) the product is a holomorphic function of the arguments which equals 1 at the origin, the logarithm of a product of exponentials of small elements of $\mathfrak{g}^c$ is given by the series of Baker, Campbell, and Hausdorff and lies in $\mathfrak{g}^c$, and the derivatives are bounded by Cauchy's estimates. For (N3), let $b_1 = \langle x, x + e_\mu\rangle$, b_2, b_3, $b_4 = \langle x, x + e_\nu\rangle$ be the bonds of p in the order $(\partial p) = b_1b_2b_3^{-1}b_4^{-1}$, $U_i \coloneqq U(b_i)$, $\alpha_i \coloneqq \alpha(b_i)$, and $F \coloneqq U(\partial p)$. Moving the factors U_i to the right we find

$$(e^{i\alpha}U)(\partial p) = e^{i\alpha_1}e^{i\beta_2}e^{-i\beta_3}e^{-i\beta_4}F\ ,$$
$$\beta_2 \coloneqq U_1\alpha_2U_1^{-1}\ , \qquad \beta_3 \coloneqq FU_4\alpha_3U_4^{-1}F^{-1}\ , \qquad \beta_4 \coloneqq F\alpha_4F^{-1}\ .$$

Here $U_1\alpha_2U_1^{-1} = \alpha_4 + (\delta_{e_\mu}\alpha)(b_4)$ and $U_4\alpha_3U_4^{-1} = \alpha_1 + (\delta_{e_\nu}\alpha)(b_1)$ by (1.1), and the conjugation by F changes an element Q by at most a constant times $|F-1||Q|$. The sum $\alpha_1 + \beta_2 - \beta_3 - \beta_4$ of the exponents is therefore at most $2d_\circ$ plus a constant times $|F-1|a_\circ$ in norm, the product of the four exponentials differs from $1 + i(\alpha_1 + \beta_2 - \beta_3 - \beta_4)$ by at most a constant times $a_\circ^2$, since its arguments are bounded by $2a_\circ$, and the multiplication by F changes norms by at most a constant factor. This gives (1.10).

Two consequences of the definition (1.9) will be used: $\Phi(X,0,0,0) = 0$ for every X and $\Phi(0,0,Z,0) = 0$ for every Z, and the linear part of Φ at the origin is $Y - A$. Hence, where the four arguments have norm at most $\rho_0/2$,

$$|\Psi(X,Y,Z,A)| \leq c_G[(|Y| + |A|)(|X| + |Y| + |Z| + |A|) + |X||Z|]\ ,$$
$$|\partial_X\Psi| \leq c_G(|Y| + |Z| + |A|)\ , \qquad |\partial_Z\Psi| \leq c_G(|X| + |Y| + |A|)\ , \tag{1.11}$$
$$|\partial_Y\Psi| \leq c_G(|X| + |Y| + |Z| + |A|)\ .$$

For the bounds of the derivatives we note that $\partial_X \Psi$ vanishes at $(X,0,0,0)$, $\partial_Z \Psi$ at $(0,0,Z,0)$, and $\partial_Y \Psi$ at the origin, and we integrate the second derivatives along the segment from these points. For the first line, $\Psi(X,0,Z,0)$ vanishes if $X=0$ or $Z=0$, so that it equals the integral of its mixed second derivative over $[0,1]^2$ along (sX,tZ) and is at most $c_G|X||Z|$; and $\Psi(X,Y,Z,A)-\Psi(X,0,Z,0)$ is the integral of the derivative in (Y,A) along (X,sY,Z,sA), which vanishes at the origin and is therefore at most $c_G(|X|+|Y|+|Z|+|A|)$.

Lemma 1.3 (The Gauge Reduction of a Slowly Varying Complex Part)**.** *Let B be a cube of the fine lattice of side $D \geq 1$ and x_0 a site of B with $x_0 + e_\mu \in B$ for every μ. Let $U = e^{i\zeta}$ on the bonds of B, with a field ζ with values in $\mathfrak{g}$, and let α be a field on the bonds of B with values in $\mathfrak{g}^c$, such that for all bonds b and unit vectors e with $b, b+e$ in B*

$$|\alpha(b)| \leq a\ , \qquad |(\delta_e \alpha)(b)| \leq a'\ , \qquad |\zeta(b)| \leq z\ , \qquad |\zeta(b+e)-\zeta(b)| \leq z'\ , \tag{1.12}$$

with $a' \leq a$ and $4aD, a, z \leq \rho_0/2$. Put

$$\omega(x) \coloneqq \sum_\mu a_\mu\,(x-x_0)_\mu\ , \qquad a_\mu \coloneqq \alpha(\langle x_0, x_0+e_\mu\rangle)\ , \qquad u_t \coloneqq e^{it\omega}\ , \tag{1.13}$$

for the sites x of B and $t \in [0,1]$. Then $u_t(x_0)=1$, $|\omega| \leq 4aD$, and $(e^{i\alpha}U)^{u_t} = e^{i\alpha_t}U$ on the bonds of B, with fields α_t with values in $\mathfrak{g}^c$ which satisfy, with $a'' \coloneqq a' + c_G z a$ and $r \coloneqq c_G[2a(4aD+2a+z)+4aDz]$,

$$|\alpha_1(b)| \leq 4Da'' + r\ , \qquad |\alpha_t(b)| \leq 2a + r\ , \tag{1.14}$$

$$|(\delta_e \alpha_t)(b)| \leq a'' + c_G z(2a+r) + c_G[(2a+z)a + (4aD+2a+z)a'' + (4aD+2a)z']\ . \tag{1.15}$$

Proof. The value $\omega(x_0)=0$ and the bound on ω are immediate from (1.13), since $|a_\mu| \leq a$ and $\sum_\mu |(x-x_0)_\mu| \leq 4D$. For a bond $b = \langle x, x+e_\mu\rangle$ of B we have $\omega(x+e_\mu) = \omega(x) + a_\mu$, and therefore

$$(e^{i\alpha}U)^{u_t}(b) = e^{it\omega(x)} e^{i\alpha(b)} e^{i\zeta(b)} e^{-i(t\omega(x)+ta_\mu)} = e^{i\alpha_t(b)} e^{i\zeta(b)}\ ,$$
$$\alpha_t(b) \coloneqq \Phi(t\omega(x), \alpha(b), \zeta(b), ta_\mu)\ ,$$

by the definition (1.9); the arguments have norm at most $\rho_0/2$ by hypothesis, and α_t has values in $\mathfrak{g}^c$ by (N2).

The plain differences of α. For $b, b+e$ in B, with $b = \langle x, x+e_\mu\rangle$, the value of U on the bond from x to $x+e$ is $e^{\pm i\zeta}$ with the value of ζ on that bond, so that by (1.1), (N1), and (1.12)

$$|\alpha(b+e)-\alpha(b)| \leq |(\delta_e\alpha)(b)| + c_G z\,|\alpha(b+e)| \leq a' + c_G z a = a''\ .$$

Two bonds of the direction μ in B are joined by a chain of at most $4D$ translations by unit vectors which stays in B, one coordinate after the other, because B is a cube. Hence $|\alpha(b) - a_\mu| \leq 4Da''$ for every bond b of the direction μ in B.

The bounds (1.14). By (N2), $\alpha_t(b) = \alpha(b) - ta_\mu + \Psi(t\omega(x), \alpha(b), \zeta(b), ta_\mu)$, and by the first line of (1.11), with $|t\omega| \leq 4aD$, the last term is at most r in norm. For $t=1$ the first two terms are at most $4Da''$ together, and for general t at most $2a$.

The bound (1.15). By (1.1) and (N1), $|(\delta_e\alpha_t)(b)| \leq |\alpha_t(b+e) - \alpha_t(b)| + c_G z(2a+r)$. The arguments of Φ at $b+e$ are $t\omega(x) + ta_e$, with $a_e \coloneqq \pm a_\nu$ for $e = \pm e_\nu$, then $\alpha(b+e)$, $\zeta(b+e)$, and the same ta_μ. The difference of the linear parts is $\alpha(b+e)-\alpha(b)$, at most a'', and by the second line of (1.11) the difference of the values of Ψ is at most $c_G(2a+z)|a_e| + c_G(4aD+2a+z)a'' + c_G(4aD+2a)z'$, the three terms being the changes of the first, the second, and the third argument, each multiplied by the bound of the corresponding derivative in (1.11). *q.e.d.*

1.3 One Carried Insertion

Let W be a group of the deviation species made at the step j. We fix a point of the space of the step $k+1$,

$$(\mathbf{U},\mathbf{J}) = (e^{i\alpha}U,\mathbf{J}) \in \mathcal{S}_{k+1}(U)\,, \tag{1.16}$$

around a real configuration U which is either U_{k+1}, for a block field V_{k+1} in the support, with the current $J \coloneqq J_{k+1}$, or the trivial configuration, with $J \coloneqq 0$; in both cases $(U,J) \in \mathcal{S}_{k+1}(U)$ and, by (b), U is one of the configurations of (1.3). By Lemma 1.1 and (a), both $(\mathbf{U},\mathbf{J})$ and (U,J) lie in $\mathcal{D}_W$. We write

$$\begin{gathered} a \coloneqq \alpha_{1,k+1}L^{-(k+1)}\,, \qquad \varrho_j \coloneqq \alpha_{1,j}L^{-j}\,, \\ D_W \coloneqq 2u_\star M\sigma(W)L^j\,, \qquad \delta_W \coloneqq D_W L^{-(k+1)} = 2u_\star M L^{-m}\sigma(W)\,. \end{gathered} \tag{1.17}$$

By (c) the ℓ^∞-diameter of X_W is at most $u_\star\sigma(W)ML^j$, and that of the dependence set X_W^+ exceeds it by at most $16M \le ML^j$; hence X_W^+ lies in a cube B_W of the fine lattice of side D_W, and δ_W is its side in the units of the scale $k+1$. The interval has at most $D_W/2$ fine bonds, and $\delta_W \ge 2ML^{-m}$ because $\sigma(W) \ge 1$.

Proposition 1.4 (One Carried Insertion)**.** *Let γ be sufficiently small. With the notation above the following hold.*

(i) For every W,

$$\|Y_W(\mathbf{U},\mathbf{J}) - Y_W(U,J)\| \le c_{\mathrm{pl}}\,\varkappa_m L^{-m}\,\|Y_W\|_{(j)}\,, \qquad c_{\mathrm{pl}} \coloneqq \max\Big(2,\ \frac{8c_G\lambda_\alpha}{c_F L}\Big)\,. \tag{1.18}$$

(ii) Let $\mu \coloneqq \min(\frac12, c_F M/(c_G\lambda_\alpha))$. If $\delta_W \le \mu$, then

$$\|Y_W(\mathbf{U},\mathbf{J})\| \le 2\|Y_W(U,J)\| + \varkappa_m L^{-m}\Big(\frac{40\,\delta_W}{\mu} + 14L^{-2m}\Big)e^{\nu\sigma(W)/8192}\,\|Y_W\|_{(j)}\,. \tag{1.19}$$

The first bound is the gain of the radii alone and holds for every group; the second has the further factor $\delta_W \le 2u_\star ML^{-m}\sigma(W)$ and is the one which is summable over the generations.

Proof. We write $r^{(n)} \coloneqq \alpha_{0,n}L^{-3n}$ for the radius of the current in (1.2) and $a_2 \coloneqq \alpha_{1,k+1}L^{-2(k+1)} = aL^{-(k+1)}$ for that of the covariant differences at the step $k+1$. By (d) and (1.5),

$$\frac{a}{\varrho_j} \le \varkappa_m L^{-m} \le \frac{1}{20}\,, \qquad \frac{a_2}{\alpha_{1,j}L^{-2j}} \le \varkappa_m L^{-2m}\,, \qquad \frac{r^{(k+1)}}{r^{(j)}} \le \varkappa_m L^{-3m} \le \frac{1}{20L^2}\,. \tag{1.20}$$

The curvature of the real configuration is $f \coloneqq \max_p|U(\partial p)-1| \le 2B_3\varepsilon_{k+1}L^{-2(k+1)}$ by (b), and $f = 0$ for the trivial configuration. Since $\varepsilon_{k+1}/\alpha_{0,k+1}$ and $\alpha_{1,n}$ tend to zero with γ, uniformly in the steps (Appendix A, $q_0 > p_0$), we may and do assume that

$$c_G f \le \tfrac14 c_F\,\alpha_{0,j}L^{-2j}\,, \qquad c_G\,\alpha_{1,n}^2 \le \tfrac14 c_F\,\alpha_{0,n}\,, \qquad \alpha_{1,n} \le \rho_0 \tag{1.21}$$

for all steps n; the first uses $\alpha_{0,k+1} \le \varkappa_m\alpha_{0,j}$ and $\varkappa_m L^{-2m} \le 1$.

(i). For $j \ge 2$ put

$$R' \coloneqq \min\Big(\frac{\varrho_j}{2a},\ \frac{c_F\,\alpha_{0,j}L^{-2j}}{8c_G\,a_2},\ \frac14\Big(\frac{r^{(j)}}{r^{(k+1)}} - 1\Big)\Big)\,,$$

and for $j = 1$ let R' be the first of the three numbers. If $2R' \le 1$, the left side of (1.18) is at most $2\|Y_W\|_{(j)} \le \|Y_W\|_{(j)}/R'$, because both points lie in $\mathcal{D}_W$. Let $2R' > 1$. For $|z| < 2R'$ the pair $(e^{iz\alpha}U, J + z(\mathbf{J} - J))$ lies in $\mathcal{S}_j(U)$: the complex part $z\alpha$ has norm less than $2R'a \le \varrho_j$ and covariant differences less than $2R'a_2 \le \varrho_j L^{-(k+1)} \le \alpha_{1,j}L^{-2j}$; by (1.10) with $a_\circ = \varrho_j \le \rho_0$ and $d_\circ = 2R'a_2$, and by (1.21) and the second entry of R', its curvature is at most $c_G f + 2c_G R'a_2 + c_G\varrho_j^2 \le \frac{3}{4}c_F\alpha_{0,j}L^{-2j}$; and the current has norm less than $r^{(k+1)} + 4R'r^{(k+1)} \le r^{(j)}$, by the third entry. The function $g(z) \coloneqq Y_W(e^{iz\alpha}U, J + z(\mathbf{J} - J)) - Y_W(U, J)$ is therefore holomorphic on the disc of radius $2R' > 1$, it is at most $2\|Y_W\|_{(j)}$ in norm there, and $g(0) = 0$, so that by Schwarz's lemma, applied to the matrix elements, $\|g(1)\| \le \|Y_W\|_{(j)}/R'$. In both cases the left side of (1.18) is at most $\|Y_W\|_{(j)}/R'$, and by (1.20) the reciprocals of the three entries of R' are at most $2\varkappa_m L^{-m}$, $(8c_G\lambda_\alpha/c_F)\varkappa_m L^{-2m}$, and $4 \cdot \frac{20}{19}\varkappa_m L^{-3m}$, each of which is at most $c_{\mathrm{pl}}\varkappa_m L^{-m}$.

(ii), the gauge. The conditions (1.2) are invariant under the G-valued gauge transformations, which conjugate α and $\mathbf{J}$, and so are the norms in (1.19), by (1.4) and because π is unitary on G. For $U = U_{k+1}$ we apply Supplement, Lemma 17.22, to a cube $\mathcal{Q}$ of 2^4 cubes of π_{k+1} which contains B_W; such a cube exists because $\delta_W \le \frac{1}{2}$ is less than the side M of these cubes. Its hypotheses hold for γ small, because the local curvature is $E \le 2B_3\varepsilon_{k+1}$ by (b) and $M_{\mathcal{Q}} \le 4M$. It gives a G-valued gauge transformation after which $U = e^{i\zeta}$ on B_W with, in fine units,

$$|\zeta(b)| \le z \coloneqq \hat{b}L^{-(k+1)}\,, \qquad |\zeta(b+e) - \zeta(b)| \le z' \coloneqq \hat{b}L^{-2(k+1)}\,, \qquad \hat{b} \coloneqq 24B_3^2 M\varepsilon_{k+1}\,,$$

and for the trivial configuration $\zeta = 0$. We work in this gauge and apply Lemma 1.3 to the cube B_W, with $D = D_W$, $x_0 = x_W$, the radii a and $a' = a_2$ of (1.16), and these z, z'; its hypotheses hold because $4aD_W = 4\alpha_{1,k+1}\delta_W \le 2\alpha_{1,k+1}$ and z are small. In the bounds of that lemma, $a_2 D_W = a\delta_W$, $zaD_W = \hat{b}\,a\delta_W$, $a^2 = \alpha_{1,k+1}L^{-(k+1)}a \le \alpha_{1,k+1}a\delta_W$, and $za = \hat{b}\,a_2$, so that every term is a multiple of $a\delta_W$, or of a, or of a_2, with a coefficient which is 4, respectively 2, respectively 1, up to terms which vanish with $\alpha_{1,k+1}$ and $\hat{b}$. For γ small we therefore have, for $t \in [0,1]$ and on the bonds of B_W,

$$|\alpha_1| \le 5a\delta_W\,, \qquad |\alpha_t| \le 3a\,, \qquad |\delta_e\alpha_t| \le 2a_2\,, \qquad |\omega| \le 4\alpha_{1,k+1}\delta_W\,. \tag{1.22}$$

(ii), the path. We claim that $(\mathbf{U}^{u_t}, R(u_t)\mathbf{J}) \in \mathcal{S}_j(U)|_{X_W^+}$ for $t \in [0,1]$. By Lemma 1.3, $\mathbf{U}^{u_t} = e^{i\alpha_t}U$, and by (1.22) and (1.20) the complex part satisfies $|\alpha_t| \le 3a < \varrho_j$ and $|\delta_e\alpha_t| \le 2a_2 \le 2\varkappa_m L^{-2m}\alpha_{1,j}L^{-2j} < \alpha_{1,j}L^{-2j}$. For the curvature we use that $\delta_W \le \mu$ and $\delta_W \ge 2ML^{-m}$ give $L^{-m} \le \mu/(2M)$, so that $2a_2 \le 2\varkappa_m L^{-m} \cdot L^{-m}\lambda_\alpha\alpha_{0,j}L^{-2j} \le (\mu\lambda_\alpha/(20M))\alpha_{0,j}L^{-2j} \le (c_F/(20c_G))\alpha_{0,j}L^{-2j}$ by the definition of μ; with (1.10) for $a_\circ = 3a \le \varrho_j$ and (1.21) the curvature is less than $c_F\alpha_{0,j}L^{-2j}$. Finally $|R(u_t)\mathbf{J}| \le (1 + c_G|\omega|)|\mathbf{J}| \le 2r^{(k+1)} < r^{(j)}$ by (N1). For $j = 1$ only the first of these conditions is posed. Since $u_1(x_W) = 1$, (1.8) and (1.7) give

$$Y_W(\mathbf{U}, \mathbf{J}) = \hat{Y}_W(\mathbf{U}^{u_1}, R(u_1)\mathbf{J})\,\pi(\mathbf{U}(I_W))\,. \tag{1.23}$$

(ii), the disc. Put

$$R_A \coloneqq \frac{\mu\varrho_j}{10\,a\delta_W}\,, \qquad R_J \coloneqq \frac{1}{3}\Big(\frac{r^{(j)}}{r^{(k+1)}} - 1\Big)\,, \qquad R \coloneqq \min(R_A, R_J)\,,$$

with $R \coloneqq R_A$ for $j = 1$, and $f(z) \coloneqq \hat{Y}_W(e^{iz\alpha_1}U, J + z(R(u_1)\mathbf{J} - J))$ for $|z| < R$, so that $f(1)$ is the first factor of (1.23) and $f(0) = \hat{Y}_W(U, J)$. Since $\delta_W \le \mu$, we have $R_A \ge \varrho_j/(10a) \ge 2$ and $R_J \ge 2$ by (1.20). For $|z| < R$ the pair in the argument of f lies in $\mathcal{S}_j(U)|_{X_W^+}$: by (1.22)

the complex part $z\alpha_1$ has norm less than $5R_A a\delta_W = \mu\varrho_j/2$; its covariant differences are less than $2R_A a_2 = \mu\varrho_j L^{-(k+1)}/(5\delta_W) \le \mu\,\alpha_{1,j}L^{-2j}/(10M)$, because $\delta_W \ge 2ML^{-m}$; by (1.10), (1.21), and the definition of μ its curvature is at most

$$c_G f + \frac{c_G\mu\lambda_\alpha}{10M}\alpha_{0,j}L^{-2j} + c_G\varrho_j^2 \le \Big(\frac14+\frac{1}{10}+\frac14\Big)c_F\,\alpha_{0,j}L^{-2j}\ ;$$

and the current has norm less than $r^{(k+1)} + 3R_J r^{(k+1)} = r^{(j)}$, because $|R(u_1)\mathbf{J}| \le 2r^{(k+1)}$. Hence f is holomorphic on the disc of radius R. Along the interval the transport of $e^{iz\alpha_1}U$ is a product of at most $D_W/2$ factors $e^{iz\alpha_1(b)}U(b)$, with $\pi(U(b))$ unitary and $\|\pi(e^{iX})\| \le e^{c'_\pi|X|}$ (Section 6.3), so that by (1.7) and (a)

$$\begin{aligned}\|f(z)\| &\le \|Y_W\|_{(j)}\exp(c'_\pi\tfrac12\mu\varrho_j\cdot\tfrac12 D_W) = \|Y_W\|_{(j)}\exp(\tfrac12 c'_\pi\mu u_\star M\alpha_{1,j}\,\sigma(W))\\ &\le e^{\nu\sigma(W)/8192}\,\|Y_W\|_{(j)}\ ,\end{aligned}$$

the last for γ so small that $4096\,c'_\pi u_\star M\alpha_{1,n} \le \nu$ for all n. By Schwarz's lemma for $f - f(0)$, which is at most twice this bound on the disc, $\|f(1)-f(0)\| \le (2/R)e^{\nu\sigma(W)/8192}\|Y_W\|_{(j)}$. In the same way $\|\pi(\mathbf{U}(I_W))\| \le \exp(c'_\pi a D_W/2) = \exp(\frac12 c'_\pi\alpha_{1,k+1}\delta_W) \le 2$, and $\|f(0)\| = \|Y_W(U,J)\|$ because $\pi(U(I_W))$ is unitary. By (1.23),

$$\|Y_W(\mathbf{U},\mathbf{J})\| \le 2(\|f(0)\| + \|f(1)-f(0)\|) \le 2\|Y_W(U,J)\| + \frac{4}{R}\,e^{\nu\sigma(W)/8192}\|Y_W\|_{(j)}\ ,$$

and $4/R \le 4/R_A + 4/R_J \le 40\varkappa_m L^{-m}\delta_W/\mu + 12\cdot\frac{20}{19}\varkappa_m L^{-3m}$ by (1.20), which is (1.19). *q.e.d.*

1.4 The Sum over the Generations

Proposition 1.5 (The Carried Groups in the Supremum Clause)**.** *Let the ensemble be dressed at the steps $1,\dots,k$, and suppose that on the term without large-field regions of the step $k+1$ the groups carried from these steps satisfy* (5.13) *and* (5.14) *at the step $k+1$, at every real background of the term, and vanish at its reference data. Let γ be sufficiently small and $L > \Lambda_{\mathfrak a} := (1+\beta_0)^{10}$. Then, for every traversal $\mathfrak n'$ of the step $k+1$,*

$$\sum_{W\ni\mathfrak n'} e^{\theta(W)+\lambda_c\nu\sigma(W)+\nu_e\,\mathrm{ext}(W)}\|Y_W\|_{(k+1)} \le 2C_\chi\tilde\rho_{k+1}\bar\delta^*_{\mathfrak n'} + \hat C\,C_{S'}\,\mathfrak a_{k+1}\ , \tag{1.24}$$

the sum over the groups of the deviation species made at the steps $j \le k$ with $\mathfrak n'$ in the interval, where $\bar\delta^*_{\mathfrak n'}$ is the expression (5.7) *of the step $k+1$ with each maximum replaced by its supremum over the real backgrounds of the term, and*

$$\hat C := \sum_{m\ge1}\Lambda_{\mathfrak a}^m\,\varkappa_m\Big[L^{-m}\Big(\frac{1280\,u_\star M}{e\,\nu\mu}+14L^{-m}\Big)+c_{\mathrm{pl}}\exp\Big(-\frac{\nu\mu L^m}{32u_\star M}\Big)\Big]\ . \tag{1.25}$$

Here $\theta(W)$, $\sigma(W)$, and $\mathrm{ext}(W)$ are the numbers fixed when W was made, as in (5.22).

The constant $\hat C$ is finite because $L > \Lambda_{\mathfrak a}$, and it depends on L, M, ν, β_0, $u_\star$, c_F, c_G, and λ_α, and not on k or on the coupling. The first term of (1.24) is at most $2C_\chi\tilde\rho_{k+1}\cdot 2c_\sigma B_3\varepsilon_{k+1}W_1$, because the bound (6.1) holds for each maximum and at every real background, and this is less than $\mathfrak a_{k+1}$ for γ small, since $\tilde\rho_{k+1}\varepsilon_{k+1} = \rho_{k+1}\varepsilon_{k+1}/\min(\alpha_{0,k+1},\alpha_{1,k+1})$ and $q_0 > p_0$. With (5.22) at the step $k+1$, in which $\ell_W \ge 1$, $S_W \ge \|Y_W\|_{(k+1)}$, and $\lambda_c \le 1$, the proposition therefore gives (5.21) at the step $k+1$ with

$$C_c := 1 + (1+\hat C)\,C_{S'}\ . \tag{1.26}$$

The constant does not come from a recursion: the bound of the carried groups uses the production clauses of the earlier steps and no supremum clause.

Proof. The amplitudes of two steps. We first show that $\mathfrak{a}_n \le \Lambda_{\mathfrak{a}} \mathfrak{a}_{n+1}$ for $1 \le n \le k$, so that $\mathfrak{a}_j \le \Lambda_{\mathfrak{a}}^m \mathfrak{a}_{k+1}$. By (3.9) and (3.10), $g_n \le (1+\beta_0) g_{n+1}$ and, for γ small, $(1+\beta_0)^{-1} r_{n+1} \le r_n \le (1+\beta_0) r_{n+1}$ for the radii $r_n = A_1 (\log g_n^{-2})^{p_0}$ of (5.6). Hence $\rho_n \le (1+\beta_0)^6 \rho_{n+1}$ and $g_n (1+r_n)^3 \le (1+\beta_0)^4 g_{n+1} (1+r_{n+1})^3$, while $e^{-c_\tau r_n^2} \le e^{-c_\tau r_{n+1}^2/(1+\beta_0)^2} \le \beta_0 g_{n+1}$ for γ small, because r_{n+1}^2 is a multiple of the power $2p_0 > 1$ of $\log g_{n+1}^{-2}$. Therefore $\tau_n \le ((1+\beta_0)^4 + \beta_0) \tau_{n+1} \le (1+\beta_0)^5 \tau_{n+1}$, and the claim follows from (7.3).

The real points. Let W be a group of the sum, made at the step j, and (1.16) a point of the space of the step $k+1$, around U. We apply Proposition 1.4(ii) if $\delta_W \le \mu$ and (i) otherwise, and in both cases

$$\|Y_W(\mathbf{U}, \mathbf{J})\| \le 2 \|Y_W(U, J)\| + \epsilon_W \ ,$$

with ϵ_W the second term of (1.19), respectively the right side of (1.18); it does not depend on the point. If U is the trivial configuration, the first term vanishes by hypothesis. If $U = U_{k+1}$, let $\mathfrak{n}_j \subset \mathfrak{n}'$ be a traversal of the step j in the interval of W and $t \coloneqq t_{\mathfrak{n}'}^{(k+1)}(W)$. As in the proof of Lemma 5.3, the comparison region of W at $\mathfrak{n}_j$ lies within the distance R_t of $\mathfrak{n}'$ in the units of the scale $k+1$, so that by (5.13) and (5.8) $e^{\theta(W)} \|Y_W(U,J)\| \le \beta_W \bar{m}_t$, where $\bar{m}_t$ is the supremum over the real backgrounds of the maximum of $|U(\partial p) - 1|$ over $p \in \mathcal{P}_t(\mathfrak{n}')$. Taking the supremum over the points we obtain

$$e^{\theta(W)} \|Y_W\|_{(k+1)} \le 2 \beta_W \, \bar{m}_{t_{\mathfrak{n}'}^{(k+1)}(W)} + e^{\theta(W)} \epsilon_W \ . \tag{1.27}$$

The numbers $\bar{m}_t$ are nondecreasing in t, and the summation by parts of the proof of Lemma 5.3, with (5.14) at the step $k+1$ and $\lambda_c \le 1$, bounds the weighted sum of the first terms of (1.27) over the groups of the sum by $2 C_\chi \tilde{\rho}_{k+1} \bar{\delta}_{\mathfrak{n}'}^*$.

The excess of one generation. Fix $j \le k$ and consider the groups of the sum which were made at the step j. Each of them has a traversal of the step j in its interval which is contained in $\mathfrak{n}'$, namely the one which contains a fine bond common to $\mathfrak{n}'$ and the interval, and $\mathfrak{n}'$ contains L^m traversals of the step j. By (5.22) at the step j, with $\ell_W \ge 1$ and $S_W \ge \|Y_W\|_{(j)}$,

$$\sum_W e^{\theta(W) + \nu \sigma(W) + \nu_e \operatorname{ext}(W)} \|Y_W\|_{(j)} \le L^m C_{S'} \, \mathfrak{a}_j \ , \tag{1.28}$$

the sum over these groups. For those with $\delta_W \le \mu$ we have $\lambda_c \nu + \nu/8192 = \frac{15}{16} \nu$ and, by (1.17) and $t e^{-\nu t/16} \le 16/(e\nu)$,

$$\begin{aligned} e^{\lambda_c \nu \sigma(W)} \epsilon_W &\le e^{\nu \sigma(W)} \varkappa_m L^{-m} \Big(\frac{80 u_\star M L^{-m}}{\mu} \sigma(W) e^{-\nu \sigma(W)/16} + 14 L^{-2m} \Big) \|Y_W\|_{(j)} \\ &\le e^{\nu \sigma(W)} \varkappa_m L^{-2m} \Big(\frac{1280 \, u_\star M}{e \, \nu \mu} + 14 L^{-m} \Big) \|Y_W\|_{(j)} \ . \end{aligned}$$

For those with $\delta_W > \mu$ we have $\sigma(W) > \mu L^m/(2 u_\star M)$, so that $e^{\lambda_c \nu \sigma(W)} \le e^{\nu \sigma(W)} e^{-\nu \sigma(W)/16} \le e^{\nu \sigma(W)} \exp(-\nu \mu L^m/(32 u_\star M))$ and

$$e^{\lambda_c \nu \sigma(W)} \epsilon_W \le e^{\nu \sigma(W)} c_{\mathrm{pl}} \, \varkappa_m L^{-m} \exp\Big(- \frac{\nu \mu L^m}{32 u_\star M} \Big) \|Y_W\|_{(j)} \ .$$

Multiplying by $e^{\theta(W) + \nu_e \operatorname{ext}(W)}$ and summing with (1.28), we find that the weighted sum of the ϵ_W over the groups made at the step j is at most the m-th term of (1.25) without the factor $\Lambda_{\mathfrak{a}}^m$, times $C_{S'} \mathfrak{a}_j$, and $\mathfrak{a}_j \le \Lambda_{\mathfrak{a}}^m \mathfrak{a}_{k+1}$.

The sum. Summing the excesses over $j \le k$, that is, over $m \ge 1$, and adding the first terms of (1.27), we obtain (1.24). *q.e.d.*

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